



The discovery of GW190521

Learn about the discovery of GW190521, the biggest gravitational wave binary observed to date. <http://digit.in/oct20-01>



Virgo and LIGO discovery

Read more about the discovery made by Virgo and LIGO – a black hole of around 142 solar masses. <http://digit.in/oct20-02>

INTERMEDIATE MASS BLACK HOLES

Astronomers have detected a class of bodies that have no business existing

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The LIGO/Virgo collaboration has announced another gravitational wave detection, the event being called GW190521. The nomenclature is based on the date of the detection, which was 21 May, 2019. The two LIGO detectors in the USA and the Virgo detector in Italy, all picked up the gravitational waves originating from the merger of two black holes. This particular detection is interesting for a number of reasons. It is the largest ever black hole detected through gravitational waves. The remnant object, which was the result of the collision, weighs 142 solar masses,

and is the first intermediate mass black hole to be detected. The two black holes that merged were 66 and 85 solar masses, and these two objects as well are in mass ranges of black holes that were never before seen. The event was so colossal that 8 solar masses were radiated as gravitational waves, which were detected by the three observatories on Earth.

When it comes to black holes, most detections of gravitational waves from black hole mergers are described as “chirps”. This one was so massive that it is being described as a bang. The signal observed was of an extremely short duration, lasting less than one tenth of a second. It originated from a source that is 17 billion light years away, in the direction of the constellation of Phoenix. However, the signal itself was formed when the universe was only half its current age. This is the farthest source from which gravitational waves have been detected by LIGO and Virgo.

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The remnant belongs to a category of elusive objects that scientists have been hunting for years. Most of the black holes that we know of are supermassive black holes, similar to Sagittarius A* at the centre of the Milky Way. Most of the supermassive black holes are also at the centre of galaxies, where there is plenty of material for them to feed on. They are millions to billions of times as massive as the Sun. The process of the formation of supermassive black holes is unclear, but it is believed to be linked to their locations in the hearts of galaxies. Scientists do not know if they are also formed from the collapse of stars, grow from stellar mass black holes that have plenty of other stars to feed on, or are formed through a previously unknown process.

The other kind of black holes are called stellar mass black holes, which are the remains of dead stars. When a star goes nova, it can outshine an entire galaxy for a few weeks, purging most of the material and energy from itself, leaving behind a core, that can implode within itself if it has sufficient mass. A stellar mass black hole has a mass equivalent to anywhere between 3 to 10 Suns. Between the stellar mass black holes and the supermassive black holes, there have been no confirmed detections. That is to say, black holes between 100 to 1000 solar masses were not detected previously. These bodies, that were hypothetical so far, are called intermediate mass black holes, and there were only a handful of candidates. The remnant of the new collision event, weighing in at 142 solar masses, is clearly an intermediate mass black hole. Astrophysicist Christopher Berry from Northwestern University says, “Long have we searched for an intermediate-mass black hole to bridge the gap between stellar-mass

Credit: Mark Myers, ABC Centre of Excellence for Gravitational Wave Discovery (OzGrav)